

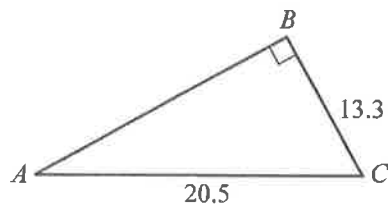
Paper 1: Basic

1. Given that $5 \sin x = 2$, find the two possible values for angle x , where $0^\circ \leq x \leq 180^\circ$.

[2]

[N22/I/19]

2.



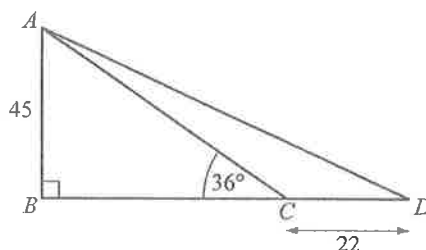
In the triangle, $AC = 20.5$ cm, $BC = 13.3$ cm and angle $ABC = 90^\circ$.

Calculate AB .

[2]

[N21/I/2]

3.



The diagram represents a tower, AB , built on horizontal ground.

The height of the tower is 45 m.

From a point C , the angle of elevation of the top of the tower is 36° .

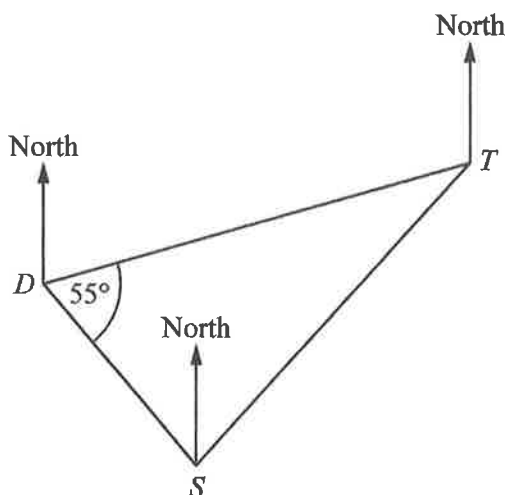
Point D is 22 m from C and BCD is a straight line.

Calculate the angle of elevation of the top of the tower from the point D .

[4]

[N20/I/18]

4.



The diagram shows the positions of three cities, Singapore (S), Delhi (D) and Tokyo (T).

Angle $SDT = 55^\circ$.

The bearing of Delhi from Singapore is 317° and the bearing of Tokyo from Singapore is 044° .

(a) Find the bearing of Singapore from Delhi.

[1]

(b) Find the bearing of Delhi from Tokyo.

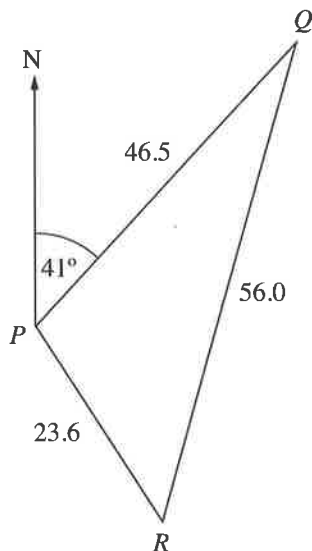
[2]

[N19/I/14]

5. The diagram shows the positions of three towns P , Q and R .

PQ is 46.5 km, PR is 23.6 km and QR is 56.0 km.

The bearing of Q from P is 041° .



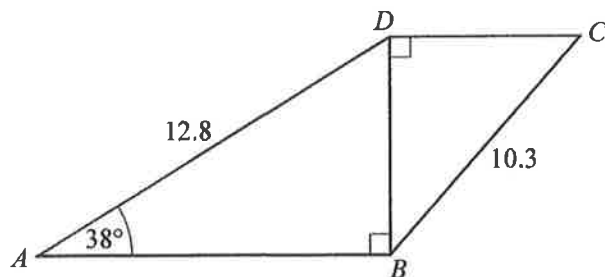
Calculate the bearing of R from P .

[4]

[N15/I/18]

Paper 1: Intermediate

6.



The diagram shows a trapezium $ABCD$.

Angle $ABD = \text{angle } BDC = 90^\circ$.

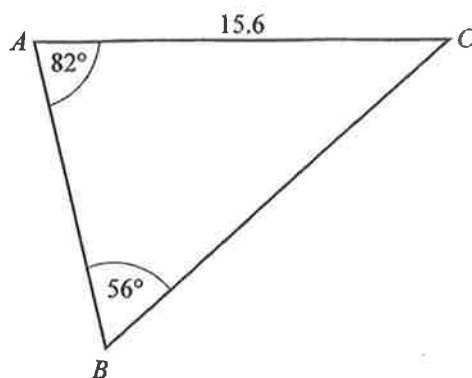
$AD = 12.8$ cm, $BC = 10.3$ cm and angle $DAB = 38^\circ$.

Calculate angle CBD .

[3]

[N22/I/22]

7.



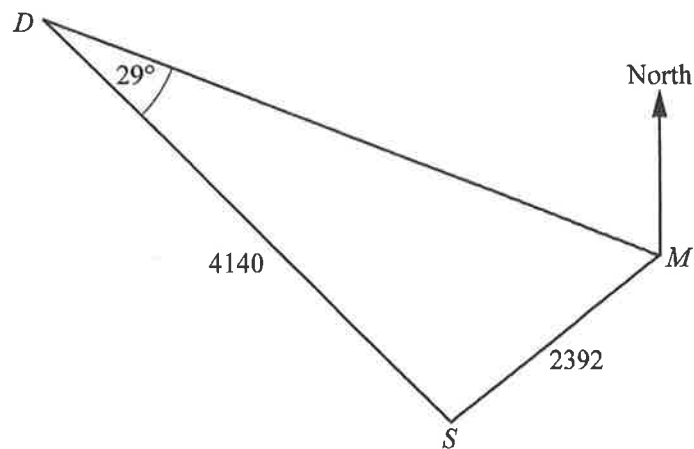
In the triangle, $AC = 15.6$ cm, angle $CAB = 82^\circ$ and angle $ABC = 56^\circ$.

Calculate the area of the triangle.

[4]

[N21/I/19]

8.



The diagram shows the positions of three cities, Singapore (S), New Delhi (D) and Manila (M). $SM = 2392$ km, $SD = 4140$ km and angle $SDM = 29^\circ$.

(a) Calculate acute angle DMS . [2]

(b) The bearing of New Delhi from Manila is 290° .
Find the bearing of New Delhi from Singapore. [2]

[N18/I/24]

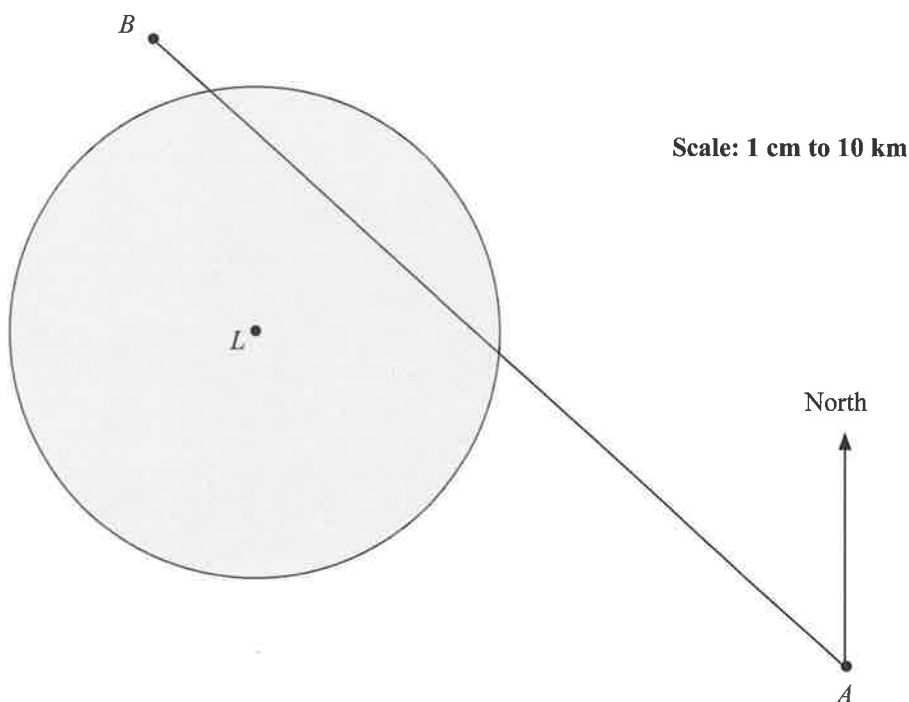
9. The area of triangle ABC is 58.6 cm^2 .

$AB = 18.7$ cm and $BC = 12.8$ cm.

Find the two possible sizes of the angle ABC . [2]

[N16/I/6]

10.



In the scale drawing, L is a lighthouse, A is a ship and B is a port.

The line AB shows the ship's course.

The light from the lighthouse is visible within the shaded area.

(a) Measure the bearing of B from A . [1]

(b) The ship sails at an average speed of 35 km/h .

Find the length of time for which the light is visible from the ship.

Give your answer in hours and minutes, correct to the nearest minute. [3]

[N16/I/22]

11.



The diagram shows a regular hexagon and an equilateral triangle.

The ratio of the perimeters hexagon : triangle = 1 : 2.

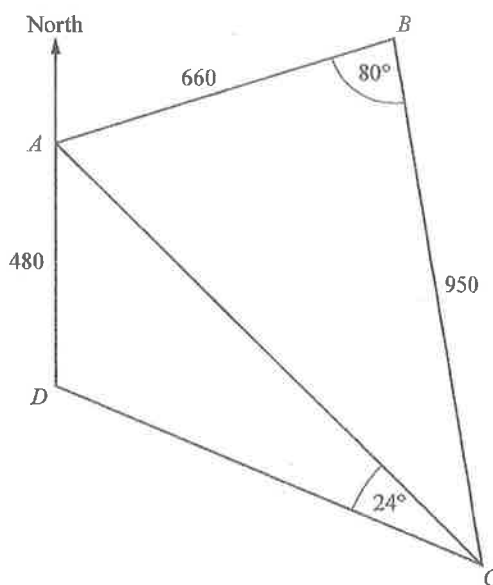
Find the ratio of the areas hexagon : triangle.

[2]

[N20/I/6]

Paper 2: Intermediate

1.



$ABCD$ is a park on horizontal ground crossed by a path AC .

A is due north of D .

$AB = 660$ m, $BC = 950$ m and $AD = 480$ m.

Angle $ABC = 80^\circ$, angle $ACD = 24^\circ$ and angle ADC is obtuse.

(a) Show that $AC = 1060$ m, correct to 3 significant figures.

[3]

(b) Calculate the bearing of D from C .

[3]

(c) Sita goes for a jog in the park at a speed of 9.5 km/h.

She starts at A , jogs around the edge to B then C , and then directly back to A .

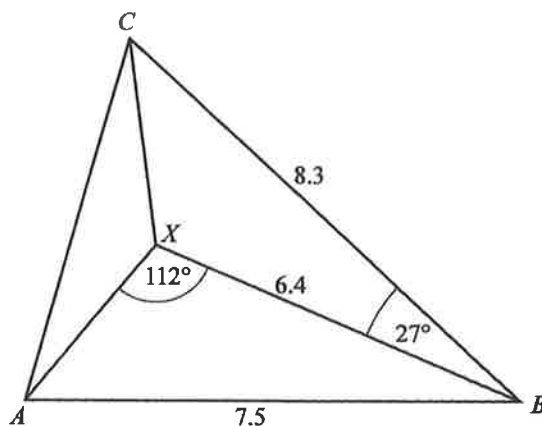
Calculate the time it takes her to jog this route.

Give your answer in minutes and seconds, correct to the nearest ten seconds.

[3]

[N20/II/4]

2.



X is a point inside triangle ABC .

$AB = 7.5$ cm, $BC = 8.3$ cm and $BX = 6.4$ cm.

Angle $AXB = 112^\circ$ and angle $XBC = 27^\circ$.

(a) Calculate CX .

[3]

(b) Calculate angle XAB .

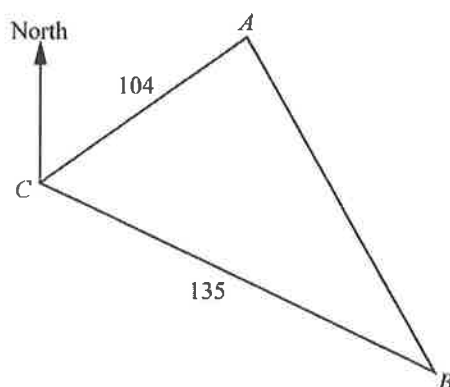
[2]

(c) Calculate the area of triangle ABC .

[3]

[N19/II/2]

3.



Points A , B and C are at sea level.

A is 104 m from C on a bearing of 064° .

B is 135 m from C on a bearing of 122° .

(a) Calculate AB .

[3]

(b) Calculate the bearing of A from B .

[3]

(c) A boat, T , travels in a straight line from C to B .

Calculate the shortest distance of T from A during this journey.

[2]

(d) C is a point at the base of a vertical cliff.

D is the point on the top of the cliff vertically above C .

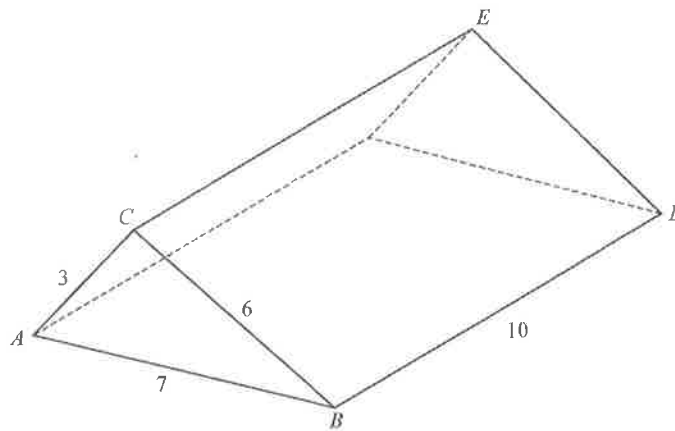
The angle of depression of B from D is 27° .

Calculate the height of the cliff.

[2]

[N17/II/5]

4.



The diagram shows a solid triangular prism with three rectangular faces.

$AB = 7$ cm, $AC = 3$ cm, $BC = 6$ cm and $BD = 10$ cm.

- Show that angle $BAC = 58.4^\circ$, correct to 1 decimal place.
- Calculate the surface area of the prism.
- Calculate the vertical distance of C above AB .
- Calculate the angle of elevation of E from A .

[3]

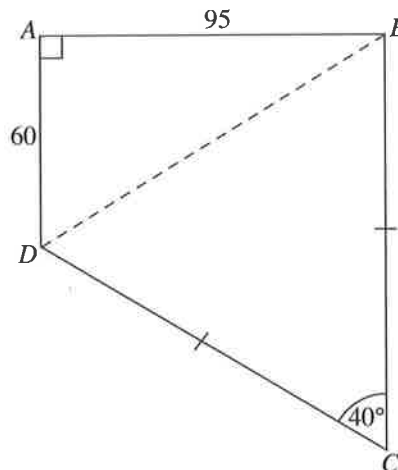
[3]

[2]

[4]

[N16/II/8]

5.



The diagram shows a field $ABCD$ on horizontal ground, crossed by a path BD .

$AB = 95$ m, $AD = 60$ m and $BC = CD$.

$\hat{B}AD = 90^\circ$ and $\hat{B}CD = 40^\circ$.

- Show that $\hat{A}DC = 127.7^\circ$, correct to one decimal place.
- Find CD .
- The land is valued at \$40 000 per hectare.
Given that 1 hectare = 10 000 square metres, calculate the value of the field.
- A bird is hovering vertically above B .
The angle of elevation of the bird from A is 18° .
Find the angle of elevation of the bird from D .

[2]

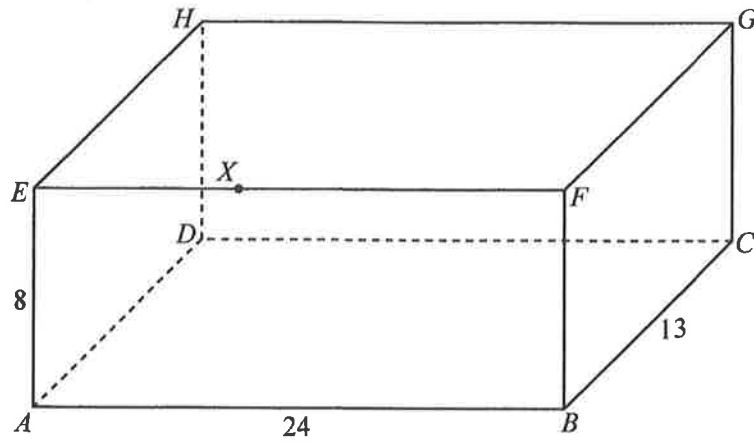
[4]

[4]

[3]

[N15/II/9]

6.



The diagram shows a cuboid $ABCDEFGH$.

$AB = 24$ cm, $BC = 13$ cm and $AE = 8$ cm.

X is a point on EF such that $EX : XF = 3 : 5$.

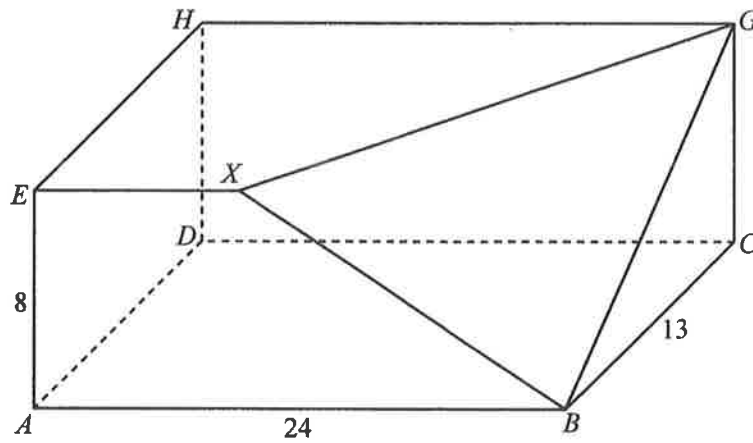
(a) Calculate the area of trapezium $ABXE$.

[3]

(b) Calculate XG .

[2]

(c)



A pyramid is cut from the cuboid as shown.

The base of the pyramid is triangle BGX .

Calculate the area of triangle BGX .

[5]

[N22/II/7]

